

Finite-state Dynamic Programming

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1 Introduction

Dynamic programming is a method for solving sequential decision problems where an economic agent makes choices over time. Rather than solving the entire problem at once, dynamic programming decomposes it into a subproblem of smaller optimization linked through the system's state. In this lecture note, we focus on **finite-state dynamic programming**, where the set of possible states is discrete and contains a finite number of elements. At each period, the agent observes the current state, receives a payoff, and the economy transitions to a new state according to a deterministic or stochastic law of motion. The objective of the economic agent is to choose a sequence of actions that maximizes the expected discounted value of current and future payoffs.

A fundamental distinction in dynamic programming is to distinguish between *state variables* and *choice variables*.

Definition 1. A *state variable* is a variable whose current value summarizes the relevant history needed to describe the agent's current situation and determine the set of feasible future allocations.

Equivalently, state variables characterize the state of the economy recursively. These typically evolve according to a law of motion and are inherited from past decisions, exogenous shocks, or aggregate conditions.

Definition 2. A *choice variable* is a variable chosen by the economic agent in a given period.

Remark 1. A variable may be a choice variable in one period and then become a state variable in the next period. Such a variable is said to be **predetermined**. For example, suppose that at date $t - 1$, the agent chooses the amount of capital to carry into date t , denoted by k_t . In other words, when date t arrives, k_t is already determined by the agent's past decision and is taken as given. Now at date t , the agent chooses k_{t+1} , the amount of capital to carry into date $t + 1$. Thus, k_t is a state variable at date t , whereas, k_{t+1} is a choice variable at date t .

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2 Robinson Crusoe Economy

We now introduce a simple Robinson Crusoe economy to illustrate finite-state dynamic programming. The Robinson Crusoe economy consists of the following environment:

- An agent called Robinson Crusoe is stranded alone on an island.
- Endowment shock:

He faces an endowment shock, such that in each period he catches a random number of fish,

$$Z_t \in \{0, \dots, B\}, Z_t \sim iid \phi$$

where Z_t is an exogenous, stochastic, independent and identically distributed process.

Remark 2. *Z_t seems to be an exogenous shock process. Then is this a state variable? No, it is not! Although Z_t affects Robinson's resources in period t , it is not a state variable in this problem. Recall that a state variable should summarize information from the past that is relevant for future decisions. Since Z_t is iid, the realization of Z_t does not depend on past realizations and does not help predict future realizations. Therefore, Z_t is simply a temporary exogenous shock that affects current resources, but it does not carry information across periods.*

- State variable and law of motion:

Let X_t denote the stock of fish available on day t . The stock evolves according to the **law of motion** such that

$$X_{t+1} = A_t + Z_{t+1}$$

- the current stock X_t is predetermined and summarizes the effects of past savings decisions A_{t-1} and exogenous shock Z_t .
- the exogenous shock is not a state variable since it is iid, i.e. the shock today gives no information about tomorrow's shock.

- Choice variable:

Given the current stock X_t , Crusoe chooses today's (1) consumption C_t and (2) savings A_t , subject to the constraint

$$A_t = X_t - C_t, 0 \leq A_t \leq M$$

Thus, C_t and A_t are the choice variable. Note that Crusoe can only store from 0 to M fish in one day so that M is the upper bound for saving.

- State space:

The state variable X_t is the sum of A_t and Z_{t+1} where each has the values of

$$A_t \in \{0, \dots, M\} \text{ and } Z_t \in \{0, \dots, B\}$$

Therefore, the maximum feasible stock is $M + B$, leading to the state space being

$$S = \{0, \dots, M + B\}$$

- Policy function:

Since the current stock X_t summarizes all relevant information, it is natural to restrict attention to strategies that depend only on the current state.¹ A **policy function** specifies Crusoe's consumption and savings choice for every possible realization of stock such that

$$C_t = \sigma_C(X_t) \quad \text{and} \quad A_t = \sigma_A(X_t).$$

The formal definition of a policy function is as follows.

Definition 3. A *policy function* is a mapping from the state space into the set of feasible choices.

Remark 3. A *policy function* is simply a decision rule assigning a feasible action to every state. At this stage, there is no requirement that the policy be optimal. Indeed, there are generally many feasible policy functions. The purpose of dynamic programming is precisely to determine which policy function maximizes the agent's lifetime utility.

- once a policy function is specified, the law of motion becomes

$$X_{t+1} = \sigma(X_t) + Z_{t+1}$$

- Objective function:

Crusoe values consumption of fish with a concave periodic (instantaneous) utility function $u(C_t)$ and discounts future at a constant daily rate ρ . Then Crusoe's lifetime utility function is the infinite sum of discounted periodic utility functions such that

$$U(C_t) = \sum_{t=0}^{\infty} \rho^t \cdot u(C_t)$$

3 Sequential to Recursive Formulation

Given the economic environment described above, Crusoe chooses his consumption and savings to maximize his expected lifetime utility subject to the constraints of the economy. In other words,

¹Such strategies are called *Markovian strategy*. Thus, we can link the finite-state dynamic programming problem with Markov chains. This will be later explained in details.

Crusoe's utility maximization problem can be expressed as the following sequence problem:²

$$\begin{aligned} \max_{\{C_t, A_t\}_{t=0}^{\infty}} \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \rho^t \cdot u(C_t) \right] \text{ subject to } A_t &= X_t - C_t, \\ X_{t+1} &= A_t + Z_{t+1}, \\ C_t &\geq 0 \text{ and } 0 \leq A_t \leq M, \\ X_0 &\text{ given} \end{aligned}$$

Note that solving directly for the entire sequence $\{C_t, A_t\}_{t=0}^{\infty}$ can be difficult because not only does are there two choice variables, but the agent must also choose infinitely many variables. Dynamic programming simplifies this infinite sequence problem by exploiting the fact that the current state summarizes all relevant information for future decisions, i.e. the infinite sequence problem can be rewritten as a recursive today-and-tomorrow problem.

To reduce the number of choice variables, we use the first constraint

$$A_t = X_t - C_t \Rightarrow C_t = X_t - A_t.$$

Then we can substitute $C_t = X_t - A_t$ into the objective function, so that savings A_t becomes the only choice variable. Next, we separate today's utility from future utility as follows:

$$\begin{aligned} \max_{\{A_t\}_{t=0}^{\infty}} \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \rho^t \cdot u(X_t - A_t) \right] &= \max_{\{A_t\}_{t=0}^{\infty}} \left\{ u(X_0 - A_0) + \rho \cdot \mathbb{E}_0 \left[\sum_{t=1}^{\infty} \rho^{t-1} u(X_t - A_t) \right] \right\} \\ &= \max_{\{A_t\}_{t=0}^{\infty}} \left\{ u(X_0 - A_0) + \rho \cdot \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \rho^t u(X_{t+1} - A_{t+1}) \right] \right\} \\ &= \max_{A_0} \left\{ u(X_0 - A_0) + \rho \cdot \mathbb{E}_0 \left[\max_{\{A_t\}_{t=1}^{\infty}} \mathbb{E}_1 \left[\sum_{t=0}^{\infty} \rho^t u(X_{t+1} - A_{t+1}) | X_1 \right] \right] \right\} \end{aligned}$$

Here, $\mathbb{E}_0$ denotes expectation using the information available today (period 0). The inner conditional expectation $\mathbb{E}_1[\cdot | X_1]$ is taken tomorrow (at period 1), given that tomorrow's stock has been realized. The inner maximization therefore represents the optimization problem that Crusoe faces tomorrow after today's savings decisions has been made. Since this optimization problem has the same structure as today's problem, it is useful to summarize it by a single function.

Up to this point, we have used the notation X_0 and X_1 to distinguish today's stock from tomorrow's stock. From now on, we adopt the standard recursive notation and denote the current state by X and the next-period state by X' . Since the same optimization problem is solved in every period, the recursive formulation focuses on the current state rather than the calendar date.

²Once X_0 is given, the agent chooses C_0 and A_0 at period 0 since $X_0 = C_0 + A_0$. Then at the beginning of period 1, Z_1 is realized, determining $X_1 = A_0 + Z_1$. Thus, A_0 is an endogenous period 0 choice, while X_0 being given is the appropriate initial condition for the infinite sequence problem.

Define

$$v(X) = \max_{\{A_t\}_{t=0}^{\infty}} \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \rho^t \cdot u(X_t - A_t) | X_0 = X \right]$$

which gives the maximum expected lifetime utility attainable when the current stock is X . Also, let

$$v(X') = \max_{\{A_t\}_{t=1}^{\infty}} \mathbb{E}_1 \left[\sum_{t=0}^{\infty} \rho^t \cdot u(X_{t+1} - A_{t+1}) | X_1 = X' \right]$$

Notice that the definition of $v(X')$ is identical to that of $v(X)$. The only difference is that the initial stock changes from X to X' . Since the economic environment is the same in every period, the optimization problem that Crusoe faces tomorrow is identical to the one he faces today except for the initial stock. This observation is known as **Bellman's Principle of Optimality**. Formally, it means that if a complete plan $\{A_t\}_{t=0}^{\infty}$ is optimal today, then once today's decision has been made and tomorrow's stock as been realized, the remaining decisions ($\{A_t\}_{t=1}^{\infty}$) must also be optimal for the problem beginning tomorrow. Therefore, tomorrow's optimization problem can be replaced by the function $v(X')$. and the recursive problem can be written as

$$v(X) = \max_A [u(X - A) + \rho \mathbb{E}_0[v(X') | X, A]] \text{ s.t. } X' = A + Z'$$

$X \text{ given}$

Since tomorrow's stock satisfies

$$X' = A + Z'$$

by the second constraint, the recursive problem can be written as

$$v(X) = \max_A u(X - A) + \rho \cdot \mathbb{E}_0[v(A + Z') | X, A]$$

For a given current stock $X = x$ and a savings choice $A = a$, the above can be expressed as

$$v(x) = \max_a \{u(x - a) + \rho \cdot \mathbb{E}_0[v(a + Z')]\}$$

The above equation expresses the optimization problem in recursive today-and-tomorrow form. This recursive formulation is the central idea of dynamic programming and is known as the **Bellman equation**.

Recall that if the current stock is x , Crusoe cannot save more than he currently has and cannot exceed the storage capacity M . Therefore, the feasible savings choices satisfy

$$0 \leq a \leq \min \{x, M\}.$$

We denote this feasible set by

$$\Gamma(x) = \{0, \dots, \min \{x, M\}\}$$

Since we are in a discrete state space,

$$\mathbb{E}_0[v(a + Z')] = \sum_{z=0}^B v(a + z) \cdot \phi(z)$$

the Bellman equation for all x becomes

$$v(x) = \max_{a \in \Gamma(x)} \left\{ u(x - a) + \rho \cdot \sum_{z=0}^B v(a + z) \cdot \phi(z) \right\} \quad (1)$$

4 Solving the Bellman Equation

Until now, we have transformed Crusoe's infinite horizon sequential optimization problem into a recursive Bellman equation. While the Bellman equation characterizes the optimal decision problem recursively, it does not directly tell us how to compute the solution. We now discuss how the Bellman equation can be solved. In doing so, we will obtain both the optimal value function satisfying the Bellman equation and the associated optimal policy function (savings rule in the Robinson Crusoe case).

Recall that the Bellman equation is

$$v(x) = \max_{a \in \Gamma(x)} \left\{ u(x - a) + \rho \cdot \sum_{z=0}^B v(a + z) \cdot \phi(z) \right\}$$

The Bellman equation contains an unknown function v . Since this function appears on both sides of the equation, it generally cannot be solved by direct algebraic manipulation. Instead, the idea of dynamic programming is to begin with an arbitrary guess of the unknown function v and improve this guess iteratively until it satisfies the Bellman equation. This improvement is guaranteed by Banach's fixed point theorem, which will be later explained in detail.

To distinguish an arbitrary guess from the unknown function v , let

$$w : S \rightarrow \mathbb{R}$$

be a candidate function. The function w need not satisfy the Bellman equation and therefore is not necessarily the solution. Rather, it serves as an approximation to the unknown function v .

Replacing the unknown function v in the Bellman equation by the candidate function w gives

$$w(x) = \max_{a \in \Gamma(x)} \left\{ u(x - a) + \rho \cdot \sum_{z=0}^B w(a + z) \cdot \phi(z) \right\}$$

Notice that the maximization determines one optimal savings decision for every state $x \in S$. Collecting these maximizing savings decisions over all states defines a feasible policy function.

Since this policy is obtained using the candidate function w , it is called a **w-greedy policy**.³

Definition 4. A policy is called **w-greedy** if for all $x \in S$,

$$\sigma_w(x) \in \operatorname{argmax}_{a \in \Gamma(x)} \left\{ u(x - a) + \rho \cdot \sum_{z=0}^B w(a + z) \cdot \phi(z) \right\}$$

for any function $w : S \rightarrow \mathbb{R}$.

4.1 Value Function Iteration

As stated earlier, the unknown function v in the Bellman equation is obtained by starting with an arbitrary candidate function (denoted by w) and improving it iteratively until it satisfies the Bellman equation. We now explain this iterative improvement in detail.

The iterative improvement is performed using the **Bellman operator**. Given a candidate function, the Bellman operator applies the Bellman equation once to produce an updated candidate function. Repeatedly applying this operator generates a sequence of candidate functions that progressively approximate the unknown function satisfying the Bellman equation. This iterative procedure is called **value function iteration (VFI)**.

Formally, the Bellman operator is a mapping $T : bS \rightarrow bS$ defined by

$$(Tw)(x) = \max_{a \in \Gamma(x)} \left\{ u(x - a) + \rho \cdot \sum_{z=0}^B w(a + z) \cdot \phi(z) \right\}$$

Since value function iteration is obtained by repeatedly applying the Bellman operator, its convergence depends on the properties of this operator. These properties are established using **Banach's Fixed Point Theorem**, which guarantees the existence and uniqueness of a fixed point whenever the Bellman operator satisfies the assumptions of the theorem.

Proposition 1. *The Bellman operator of the Robinson Crusoe economy has a unique fixed point v^* .*

Proof. The Bellman operator of the Robinson Crusoe economy satisfies the following properties:

- $T : bS \rightarrow bS$ is a self-map.
- T is a uniform contraction on (bS, d_∞) where

$$d_\infty(Tw, w) = \sup_{x \in S} |(Tw)(x) - w(x)|$$

- Since T is a self-map in a complete metric space (bS, d_∞) and T is a uniform contraction on bS with modulus ρ , the conditions for Banach's fixed point theorem is satisfied. Thus, the

³Again, since the candidate function w need not equal the optimal value function v^* , a w -greedy policy need not be optimal.

unique fixed point v^* such that

$$Tv^* = v^*$$

exists.

□

Algorithm 1: Value Function Iteration (VFI)

- 1: Set numerical tolerance $\varepsilon > 0$
 - 2: Pick any $w \in bS$, i.e. an initial candidate function of v .
 - 3: Repeat
 - compute Tw using w
 - set $e = d_\infty(Tw, w)$
 - set $w = Tw$
 - 4: Until $e < \varepsilon$
-

Upon convergence, the final iterate is the unique fixed point v^* of the Bellman operator and therefore satisfies the Bellman equation. Once the optimal value function has been obtained, the corresponding policy function is recovered by selecting, for each state x , the savings action that maximizes the right-hand side of the Bellman equation.

Theorem 1. *A policy σ is optimal if and only if it is v^* -greedy.*

Thus, the optimal policy function, denoted σ^* , is obtained by evaluating the greedy policy using the fixed point value function v^* . In other words, value function iteration searches for the value function, while the optimal policy is recovered only after the value function has converged.

4.2 Policy Function Iteration

Rather than iterating on the value function, an alternative algorithm iterates directly on the policy function. This procedure is called **policy function iteration (PFI)**.

5 Non-iid Shock Case

Suppose the exogenous shock Z is now non-iid, i.e. the current shocks affect future allocations so that Z_t is also a state variable. Suppose Z is a Markov process with stochastic kernel $p_z(z, z')$. The Bellman equation is now

$$v^*(x, z) = \max_{a \in \Gamma(x)} u(x - a) + \rho \cdot \sum_{z'} v^*(a + z', z') \cdot p_z(z, z')$$

6 Stochastic Recursive Sequences (SRS)

Once the Bellman equation has been solved and the optimal policy function has been obtained, we are often interested in studying the implications of the model economy. For example, we may wish to compute the stationary distribution of the state variable or compare moments generated by the model with those observed in the data. This requires simulating sample paths of the economy over time.

Although the dynamics of the economy are characterized by a Markov chain through its transition probabilities, this representation is less convenient for numerical simulation. In practice, computers generate independent uniform random numbers rather than transition probabilities directly. It is therefore convenient to represent the Markov chain as a deterministic function of the current state and an independent random shock. This representation is called a **stochastic recursive sequence (SRS)**.

Definition 5. A *stochastic recursive sequence (SRS)* has the form

$$X_{t+1} = F(X_t, Z_{t+1}), \quad X_0 \sim \psi \in \mathcal{P}(S)$$

where $F : S \times Z \rightarrow S$ and Z_t is iid shocks.

The definition of a stochastic recursive sequence may appear unfamiliar at first. However, we have already encountered this representation in the Robinson Crusoe economy. Once the optimal policy function σ^* has been obtained after VFI, the law of motion of the economy becomes

$$X_{t+1} = \sigma^*(X_t) + Z_{t+1}$$

where $\sigma^*(X_t)$ is the optimal savings decision and Z_{t+1} is the exogenous shock. Defining

$$F(X_t, Z_{t+1}) = \sigma^*(X_t) + Z_{t+1}$$

the law of motion can be written as

$$X_{t+1} = F(X_t, Z_{t+1})$$

which is exactly the form of a stochastic recursive sequence. Thus, an SRS is not a new model, but rather is simply another way of representing the law of motion of a dynamic economy. Using this law of motion representation, we can simulate the evolution of the state variable and recover the corresponding allocations using the optimal policy function.

Remark 4. Indeed, a stochastic recursive sequence has a Markov property. Given a SRS such that

$$X_{t+1} = F(X_t, Z_{t+1})$$

notice that the shock sequence $\{Z_t\}$ is iid, i.e. the realization of tomorrow's shock is independent of

all previous shocks. Moreover, the current state X_t summarizes all relevant information inherited from the past, i.e. once the current state is known the distribution of the next state depends only on X_t . This can be written as

$$\mathbb{P}(X_{t+1}|X_t, X_{t-1}, \dots) = \mathbb{P}(X_{t+1}|X_t)$$

which means that every stochastic recursive sequence satisfies the Markov property.

We can take one step even further and show that not only every SRS satisfies the Markov property, but also every Markov chain can also be represented as a stochastic recursive sequence. Therefore, the two representations are mathematically equivalent and the choice between them is a matter of convenience rather than substance.

6.1 SRS to Markov Chain

Suppose an SRS is given by

$$X_{t+1} = F(X_t, Z_{t+1}) \text{ and } X_0 \sim \psi \in \mathcal{P}(S)$$

First, the initial distribution is given by the definition of the SRS itself such that

$$X_0 \sim \psi \in \mathcal{P}(S)$$

Also, since the shock distribution is known, the probability of moving from state x to state y is simply the probability that the shock generates y through the function F . Hence, the transition probabilities are

$$p(x, y) = \mathbb{P}[F(x, Z) = y]$$

Thus, the Markov chain generated by the SRS is completely determined by the pair (ψ, P) .

6.2 Markov Chain to SRS

We have shown that every stochastic recursive sequence induces a Markov chain. We now establish the converse result by showing that every Markov chain can be represented as a stochastic recursive sequence.

The main idea is to replace the probabilistic transition rule of the Markov chain with a deterministic function driven by an iid random shock. Rather than selecting the next state directly according to the transition probabilities, we generate an independent random variable $Z_t \sim U[0, 1]$. The unit interval is then partitioned into subintervals whose lengths coincide with the transition probabilities from the current state. The realization of the uniform shock determines which interval is selected and, consequently, the next state.

This construction can be illustrated in the following example.

Example 1. Consider a discrete Markov chain with an initial distribution $\psi \in \Delta^3$ and the following stochastic kernel

$$p = \begin{bmatrix} 0.7 & 0 & 0.3 & 0 \\ 0 & 0.6 & 0 & 0.4 \\ 0.3 & 0 & 0.7 & 0 \\ 0 & 0.1 & 0 & 0.9 \end{bmatrix}$$

Then we can generate a SRS with the same dynamic as this Markov chain as follows. The distribution of the initial state is the same as $X_0 \sim \psi$. The law of motion is given by

$$X_{t+1} = F(X_t, Z_{t+1})$$

where Z_{t+1} is an iid uniform shock. Note that

$$\begin{aligned} F(1, z) &= \begin{cases} 1 & \text{with } 0 \leq z < 0.7 \\ 3 & \text{with } 0.7 \leq z \leq 1 \end{cases} \\ F(2, z) &= \begin{cases} 2 & \text{with } 0 \leq z < 0.6 \\ 4 & \text{with } 0.6 \leq z \leq 1 \end{cases} \\ F(3, z) &= \begin{cases} 1 & \text{with } 0 \leq z < 0.3 \\ 3 & \text{with } 0.3 \leq z \leq 1 \end{cases} \\ F(4, z) &= \begin{cases} 2 & \text{with } 0 \leq z < 0.1 \\ 4 & \text{with } 0.1 \leq z \leq 1 \end{cases} \end{aligned}$$

In the example above, the intervals assigned to each possible next state have lengths equal to the corresponding transition probabilities. For instance, when the current state is 1, the interval $[0, 0.7)$ is assigned to state 1 and the interval $[0.7, 1]$ is assigned to state 3. Since a uniform random variable falls into these intervals with probabilities 0.7 and 0.3, respectively, the resulting transitions coincide exactly with the first row of the transition matrix. The same construction is applied to every current state.

This idea extends immediately to an arbitrary finite-state Markov chain. For each current state $x \in S$, partition the unit interval into subintervals

$$I(x; 1), I(x; 2), \dots, I(x; n)$$

whose lengths are given by the transition probabilities

$$p(x, x_1), p(x, x_2), \dots, p(x, x_n)$$

where $x_1, \dots, x_n$ denote the elements of the state space. The transition function is then defined by

$$F(x, z) = \sum_{i=1}^n x_i \cdot \mathbb{1}\{z \in I(x; i)\}$$

so that the realized value of the uniform shock uniquely determines the next state. Since

$$\mathbb{P}(F(x, Z) = x_i) = p(x, x_i)$$

the stochastic recursive sequence generated by F has exactly the same transition probabilities as the original Markov chain.

Therefore, given a Markov chain (P, ψ) , the equivalent stochastic recursive sequence is given by

$$\begin{aligned} X_0 &\sim \psi \in \mathcal{P}(S) \\ X_{t+1} &= F(X_t, Z_{t+1}) = \sum_{i=1}^n x_i \cdot \mathbb{1}\{Z_{t+1} \in I(X_t; i)\} \end{aligned}$$

where Z_t is an iid sequence of $U[0, 1]$ random variables and, for each current state x , the intervals $\{I(x; i)\}_{i=1}^n$ partition the unit intervals with lengths equal to the transition probabilities $\{p(x, x_i)\}_{i=1}^n$.

7 Coupling and Stability

Once a stochastic recursive sequence has been constructed, we can simulate the evolution of the economy over time. A natural question is whether the long-run behavior of the economy depends on the initial state. In other words, if two economies begin from different initial conditions but evolve according to the same law of motion, will they eventually exhibit the same long-run behavior?

This question is referred to as **global stability**. One powerful technique for establishing global stability is **coupling**.⁴

7.1 Coupling

The basic idea of **coupling** is to compare two copies of the same stochastic process. Suppose one economy begins from state $X_0 = x$, while another begins from $X'_0 = x'$. Both evolve according to the same transition kernel P . If we can construct the two processes so that they eventually meet and thereafter remain together, then the long-run behavior of the economy is independent of the initial state.

⁴A useful summary is that there are three ways to verify whether a Markov chain is globally stable. A Markov chain is globally stable if (1) The Dobrushin coefficient $\alpha(p^t) > 0$ for some t , (2) the Markov chain is irreducible and aperiodic, and (3) coupling.

To formalize this idea, suppose that

$$(X_t)_{t \geq 0} \text{ and } (X'_t)_{t \geq 0}$$

are two Markov chains with the same transition kernel P , but possibly different in states such as

$$X_0 = x, \quad X'_0 = x'$$

Consider the joint process

$$(X_t, X'_t)_{t \geq 0}$$

Since the two chains evolve independently according to the same transition kernel, the transition probability of the joint process is given by

$$P\{X_{t+1} = y, X'_{t+1} = y' \mid X_t = x, X'_t = x'\} = p(x, y)p(x', y')$$

A natural quantity to study is the probability that the two chains meet after one transition. Given the current states $X_t = x$ and $X'_t = x'$, this probability is

$$g(x, x') = P\{X_{t+1} = X'_{t+1} \mid X_t = x, X'_t = x'\}$$

Since the two chains can meet at any state $y \in S$, we have

$$g(x, x') = \sum_{y \in S} p(x, y)p(x', y)$$

The quantity $g(x, x')$ depends on the current states x and x' . To obtain a uniform lower bound that holds regardless of the initial states, define

$$\varepsilon = \min_{x, x' \in S} g(x, x')$$

If $\varepsilon > 0$, then no matter where the two chains begin, there is always at least probability ε that they meet after one transition. Once the two chains meet, we can construct them to evolve together thereafter. Consequently, repeated application of this argument implies that the effect of the initial condition gradually disappears.

This observation leads to the following result.

Lemma 1. *Suppose that $\varepsilon > 0$. Then the Markov chain is globally stable.*

Although coupling provides an intuitive criterion for global stability, the quantity of ε is often difficult to verify directly. A more convenient condition is by the Dobrushin coefficient, which measures the similarity of the transition probabilities from different initial states.

The Dobrushin coefficient provides a practical criterion for establishing global stability, whereas coupling provides an intuitive explanation of why global stability holds. The key idea is that if two

copies of the same Markov chain can be constructed so that they eventually meet and thereafter evolve together, then the stochastic process eventually forgets its initial condition. Consequently, regardless of the starting state, the economy converges to the same long-run behavior.

This concludes our study of finite-state dynamic programming. We began by formulating the agent's optimization problem recursively through the Bellman equation and obtained an optimal policy function. The optimal policy induces a stochastic recursive sequence that describes the evolution of the state variable. Finally, coupling provides an intuitive way to understand why this stochastic process is globally stable and why its long-run behavior does not depend on the initial state.

A Expressing the Objective and Policy Function using Markov Operators

A.1 Policy Function and Markov Operator

Recall that a policy function specifies the feasible choice as a function of the current state. In the Robinson Crusoe economy, the policy function determines the optimal savings decision according to

$$A_t = \sigma(X_t)$$

Equivalently, the policy function is a mapping $\sigma : S \rightarrow \{0, \dots, M\} \subseteq S$ such that

$$\sigma(x) = \begin{bmatrix} A(X_t = 0) \\ \vdots \\ A(X_t = x) \\ \vdots \\ A(X_t = M + B) \end{bmatrix}_{(M+B+1) \times 1} \in \Sigma$$

where Σ is the set of feasible policy functions. Thus, for every possible stock level, the policy function prescribes the amount of fish that should be saved.

Once a policy function has been specified, the agent's behavior is completely determined. Consequently, the evolution of the state variable is governed by the law of motion

$$X_{t+1} = \sigma(X_t) + Z_{t+1}$$

Since the shock sequence Z_t is iid, the distribution of tomorrow's state depends only on the current state. Hence,

$$P(X_{t+1} = y \mid X_t = x, X_{t-1}, \dots) = P(X_{t+1} = y \mid X_t = x).$$

so that the stochastic process $\{X_t\}_{t \geq 0}$ satisfies the Markov property. Therefore, every feasible policy function induces a Markov chain in the state space S .

For each current state $x \in S$, the policy function prescribes a savings decision $\sigma(x)$. Let $\gamma(\sigma(x), y)$ denote the probability that tomorrow's stock is y when today's savings is $\sigma(x)$. Then the stochastic kernel of the Markov chain, induced by the policy function, is therefore

$$p_\sigma(x, y) = \gamma(\sigma(x), y)$$

which gives the probability of moving from the current state x to the next-period state y under the policy function σ .

Associated with the stochastic kernel p_σ is the Markov operator M_σ , defined by

$$(M_\sigma^t \cdot h)(x) = \sum_{y \in S} p_\sigma^t(x, y) \cdot h(y), \quad t \geq 0$$

for every bounded function $h : S \rightarrow \mathbb{R}$. Since the Markov operator computes conditional expectations of future functions of the state variable, the above can be interpreted as the *expectation of t -period objects $(h(X_t))$ given the initial shock $X_0 = x$* , i.e.

$$\mathbb{E}[h(X_t)|X_0 = x]$$

A.2 Objective Function and Markov Operator

The Markov operator introduced in the previous lecture notes provides an alternative representation of the objective function associated with a policy function. Recall that, for a given feasible policy $\sigma \in \Sigma$, the objective function is

$$v_\sigma(x) = \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \rho^t \cdot u(X_t - \sigma(X_t)) \right]$$

where $\mathbb{E}_0$ denotes the expectation taken at period 0, conditional on the initial state $X_0 = x$. Since the future evolution of the state variable depends on the random shocks, this expectation averages over all possible future paths of the state process generated by the policy function. Our goal is to express this expectation using the Markov operator.

To understand how this representation arises, it is useful to first consider a finite-horizon problem. Over a horizon of length T , the expected discounted utility is

$$\mathbb{E}_0 \left[\sum_{t=0}^T \rho^t \cdot u(X_t - \sigma(X_t)) \right]$$

One way to evaluate this expectation is to enumerate every possible trajectory of the state variable. Let $x = (x_0, \dots, x_T) \in S^{T+1}$ denote a possible realization of the state process. The total discounted

utility generated by this trajectory is

$$F(x) = \sum_{t=0}^T \rho^t \cdot u(x_t - \sigma(x_t))$$

while

$$q_{T+1}(x) = \mathbb{P}(X_0 = x_0, \dots, X_T = x_T)$$

denotes the joint probability that this trajectory occurs. Therefore,

$$\mathbb{E} \left[\sum_{t=0}^T \rho^t \cdot u(X_t - \sigma(X_t)) \right] = \sum_{x \in S^{T+1}} F(x) \cdot q_{T+1}(x) \quad (2)$$

Although equation 2 is mathematically correct, it becomes increasingly difficult to evaluate as the horizon grows. As T increases, the number of possible trajectories expand rapidly, making direct computation impractical.⁵

A more convenient representation follows from the linearity of expectation

$$\mathbb{E}_0 \left[\sum_{t=0}^T \rho^t \cdot u(X_t - \sigma(X_t)) \right] = \sum_{t=0}^T \rho^t \cdot \mathbb{E}_0 [u(X_t - \sigma(X_t))]$$

Define the one-period utility function

$$r_\sigma(X_t) \equiv u(X_t - \sigma(X_t)),$$

the above can be written as

$$\mathbb{E}_0 \left[\sum_{t=0}^T \rho^t \cdot u(X_t - \sigma(X_t)) \right] = \sum_{t=0}^T \rho^t \cdot \mathbb{E}_0 r_\sigma(X_t)$$

Now since the rightward operation of the Markov operator can be interpreted as an expectation, we get

$$\begin{aligned} \sum_{t=0}^T \rho^t \cdot \mathbb{E}_0 r_\sigma(X_t) &= \sum_{t=0}^T \rho^t \sum_{y \in S} p_\sigma^t(x, y) r_\sigma(y) \\ &= \sum_{t=0}^T \rho^t \cdot M_\sigma^t \cdot r_\sigma(x) \\ &= \underbrace{\rho^0 M_\sigma^0 r_\sigma(X_0)}_{\text{average of time } t=0 \text{ when initial is } X_0} + \dots + \underbrace{\rho^T M_\sigma^T r_\sigma(X_0)}_{\text{average of time } t=T \text{ when initial is } X_0} \end{aligned}$$

⁵As the horizon increases, the set of possible trajectories becomes S^{T+1} , whose cardinality is $|S|^{T+1}$. For example, $|S|(ift=0) \rightarrow |S \times S| = (M+B+1)^2(ift=1) \rightarrow \dots$

The interpretation is straightforward. The quantity

$$(M_{\sigma}^t r_{\sigma})(x)$$

is the expected period utility received at period t , conditional on the initial state being $X_0 = x$. Thus, each iterate of the Markov operator computes the expected contribution of one future period, while the discounted sum aggregates these contributions over time.

Finally, the same argument holds as $T \rightarrow \infty$. Thus, we can recover our finite-horizon into an infinite horizon problem such that

$$v_{\sigma}(x) \equiv \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \rho^t \cdot u(X_t - \sigma(X_t)) \right] = \sum_{t=0}^{\infty} \rho^t M_{\sigma}^t r_{\sigma}(X_0 = x)$$

Therefore, the objective function associated with any feasible policy can be written entirely in terms of the Markov operator. Rather than computing expectations over every possible trajectory explicitly, the Markov operator uses the transition probabilities of the induced Markov chain to compute the corresponding conditional expectations.